

Quantification: 542_z06

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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
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lacuna count	16
interior lacuna count	13
frame-edge lacuna count	3
gap bridges added (detection)	17

Interior lacunae only: mean, median and sample SD (n = 13)

measure	kind	mean	median	sd	units
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area_px2	lacuna shape	2541.77	3013	1427.44	px^2
major_axis_length_px	lacuna shape	108.237	124.464	40.2415	px
minor_axis_length_px	lacuna shape	31.7238	33.1799	10.2747	px
aspect_ratio	lacuna shape	3.5771	3.6285	1.3089	unitless
eccentricity	lacuna shape	0.935	0.9613	0.0596	unitless
solidity	lacuna shape	0.7472	0.7672	0.1322	unitless
roots_count	headline	8.6154	8	3.305	count
ring_length_r30_px	headline	361.462	319	152.835	px
ring_length_r60_px	reported	907.385	816	333.2	px
owned_length_px	ownership-dependent	988.281	919.293	748.381	px
edge_count	ownership-dependent	30.8462	20	21.0113	count
mean_edge_length_px	ownership-dependent	31.5991	33.5078	7.5869	px
perimeter_px	normalised	297.812	309.522	115.117	px
ring_area_r30_px2	normalised	10771.7	11337	2967.43	px^2
ring_area_r60_px2	normalised	25565.5	27170	6351.05	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9908	1	0.0272	unitless
ring_density_r30	normalised	0.0327553	0.0337832	0.00638738	px^-1
ring_density_r60	normalised	0.0350223	0.0338849	0.00605335	px^-1
roots_per_100px_perimeter	normalised	2.9437	3.0734	0.4703	per 100 px
ring_attached_length_r30_px	network v2	273.308	276	139.079	px
ring_attached_length_r60_px	network v2	501	371	292.662	px
ring_length_w_r30_px	network v2	408.263	353.806	174.91	px
ring_length_w_r60_px	network v2	1023.96	920.193	381.461	px
sholl_crossings_r10	network v2	9.7692	10	4.5854	count
sholl_crossings_r20	network v2	12.5385	11	5.1416	count
sholl_crossings_r30	network v2	14	15	5.6421	count
ring_width_mean_r30_px	width	5.7445	5.7752	0.2709	px

Per field

measure	value	units
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analysed_area_px2	1.00906e+06	px^2
total_skeleton_length_px	34964	px
canalicular_length_density_per_px	0.03465	px^-1
junction_count	375	count
junction_density_per_px2	0.00037163	px^-2
lacuna_count	16	count
lacunae_per_px2	1.586e-05	px^-2

skeleton_component_count	654	count
mean_component_length_px	53.4618	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0390336	px^-1
field_density_without_flagged_per_px	0.0345162	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	6.1924	px
width_median_px	6.3246	px
width_p10_px	4	px
width_p90_px	8	px
width_pixels_used	33745	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	445	56.9837	11.5357	4.9398	0.9793	0.7672	-0.2356	148.654	554.791	False
3	556	44.867	25.9744	1.7274	0.8154	0.5265	-0.4071	66.9281	106.103	False
4	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
5	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
6	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
7	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
8	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
9	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
10	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
11	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
12	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
13	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
14	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
15	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
16	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	176	516	1032.99	33	31.3028	6.3269
2	4	177	487	335.078	10	33.5078	5.7426
3	5	218	612	292.936	15	19.5291	5.4158
4	5	200	560	387.191	14	27.6565	5.467
5	7	264	816	504.931	15	33.6621	5.8182
6	11	665	1478	1337.61	47	28.4599	5.6931
7	8	305	727	688.659	19	36.2452	5.8374
8	10	319	663	414.12	20	20.706	5.9249
9	10	391	962	1157.4	33	35.0728	6.3867
10	8	383	1079	919.293	20	45.9647	5.7963
11	13	606	1550	1771.6	55	32.211	5.7752

12	13	449	1081	955.24	42	22.7438	5.5623
13	13	471	1041	2992.18	83	36.0504	5.9217
14	5	251	740	1091.4	28	38.9786	5.3371
15	4	264	763	404.392	20	20.2196	5.4115
16	4	114	467	97.0833	6	16.1805	4.3124

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0313334	0.0406491	2.7877
2	126.468	6226	14794	1	1	0.0284292	0.0329187	3.1629
3	160.64	6976	18799	1	0.9774	0.03125	0.0325549	3.1126
4	151.782	6752	15321	1	1	0.0296209	0.0365511	3.2942
5	322.043	11663	28685	1	1	0.0226357	0.0284469	2.1736
6	438.517	13901	31616	1	1	0.0478383	0.0467485	2.5085
7	211.924	8845	21455	1	1	0.0344828	0.0338849	3.7749
8	359.191	12383	27034	1	0.9025	0.0257611	0.0245247	2.784
9	302.451	11295	27170	1	1	0.0346171	0.0354067	3.3063
10	309.522	11337	28286	1	1	0.0337832	0.0381461	2.5846
11	468.274	15363	34096	1	1	0.0394454	0.0454599	2.7762
12	422.978	13165	30001	1	1	0.0341056	0.0360321	3.0734
13	379.848	13296	32108	1	1	0.0354242	0.0324218	3.4224
14	217.924	8830	22986	1	1	0.0284258	0.0321935	2.2944
15	332.894	10505	23806	0.8602	0.7919	0.0251309	0.0320507	1.2016
16	205.776	6766	16357	0.7841	0.714	0.016849	0.0285505	1.9439

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	176	382	200.267	588.316	6	5	6
2	106	202	202.894	542.345	4	7	10
3	136	185	246.794	688.227	4	8	6
4	149	207	221.924	623.316	5	7	6
5	172	304	294.35	920.193	8	11	13
6	532	1012	758.311	1693.23	18	20	22
7	204	332	343.291	824.039	8	9	14
8	276	371	353.806	741.34	10	10	8
9	313	483	439.647	1078.76	13	15	16
10	309	705	433.747	1229.71	10	13	15
11	480	1020	684.985	1745.91	15	23	25
12	331	663	504.931	1203.08	14	13	15
13	412	700	541.529	1188.76	13	18	17
14	133	329	281.208	832.524	5	9	15
15	140	284	286.037	848.583	6	11	14
16	86	89	126.882	516.961	3	3	6

